

"Technical Specifications & Constraints"

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1. Technical Overview

IdeaDNA is a full-stack web application built on modern cloud-native architecture. The platform requires specific technical capabilities across frontend, backend, database, AI services, and infrastructure layers. This document defines all technical requirements, constraints, and specifications for successful implementation.

Architecture Pattern: Serverless frontend (Next.js on Vercel) + Serverless backend (API Routes) + Managed database (PostgreSQL) + External AI services (OpenAI). All components communicate via HTTPS.

2. Hardware Requirements

2.1 Development Environment

- CPU: Intel i5 / AMD Ryzen 5 or higher (4+ cores)
- RAM: 16GB minimum, 32GB recommended
- Storage: 50GB SSD for project files, dependencies, and Docker images
- Network: Stable broadband connection (10+ Mbps) for API calls
- OS: macOS 13+, Ubuntu 22.04+, or Windows 11 with WSL2

2.2 Production Environment (Serverless)

- Frontend: Vercel Edge Network (no dedicated hardware needed)
- Backend: Vercel Serverless Functions (auto-scaling, 1024MB-3008MB RAM per function)
- Database: Railway/Supabase managed PostgreSQL (shared CPU, 1-4GB RAM)
- Cache: Upstash Redis (serverless, auto-scaling)
- AI API: OpenAI managed infrastructure

3. Software Requirements

3.1 Runtime Requirements

Component	Requirement	Version	Purpose
Node.js	JavaScript runtime	18.x LTS or 20.x LTS	Next.js server runtime
npm	Package manager	9.x+	Dependency management
TypeScript	Type system	5.3+	Type safety
PostgreSQL	Database	15.x	Primary data store
Redis	Cache store	7.x	Session + response caching
Git	Version control	2.40+	Source code management

3.2 Framework & Library Requirements

Category	Technology	Version	Purpose
Framework	Next.js	14.x	React framework with App Router
UI	React	18.x	Component library
Styling	Tailwind CSS	3.4+	Utility-first CSS
Animation	Framer Motion	11.x	UI animations
State	Zustand	4.5+	Client state management
Server State	React Query (TanStack)	5.x	Server state + caching
ORM	Prisma	5.10+	Database ORM
Auth	Clerk	Latest	Authentication SDK
AI	LangChain	0.1.x	LLM orchestration
AI	OpenAI SDK	4.x	GPT-4 API client
Charts	D3.js	7.x	Radar charts, data viz
Graphs	React Flow	12.x	Evolution trees
Validation	Zod	3.22+	Schema validation
PDF	PDFKit / jsPDF	Latest	PDF export
Testing	Playwright	1.40+	E2E testing
Testing	Vitest	1.x	Unit testing

4. Network Requirements

Protocol

HTTPS required for all communications. TLS 1.3 preferred, TLS 1.2 minimum.

CORS

Whitelist-only origins. No wildcard (*) in production. localhost:3000 for development.

Rate Limiting

IP-based: 100 req/min. User-based: tier-dependent (see Security Document).

WebSocket

Not required for MVP. Future: real-time collaboration may need WebSocket/SSE.

CDN

Vercel Edge Network for static assets. Global distribution, automatic caching.

DNS

Custom domain with Cloudflare DNS. A/AAAA records pointing to Vercel.

5. Database Requirements

5.1 PostgreSQL Configuration

- Version: PostgreSQL 15.x or higher
- Extensions: pgvector (for embeddings), uuid-oss (for UUID generation)
- Connection Pool: PgBouncer with 20 max connections
- Backup: Automated daily backups, 30-day retention
- SSL: Required for all connections
- Character Set: UTF-8
- Timezone: UTC for all timestamps

5.2 Redis Configuration

- Version: Redis 7.x
- Max Memory: 512MB for MVP, scalable
- Eviction Policy: allkeys-lru (least recently used)
- Persistence: AOF every second
- SSL: TLS encryption for connections

6. AI/ML Requirements

6.1 OpenAI API Requirements

- Model: GPT-4 (gpt-4-turbo-preview) for generation tasks
- Embeddings: text-embedding-3-small for semantic search
- Context Window: 128K tokens for GPT-4
- Temperature: Agent-specific (0.2 - 0.6)
- Max Tokens: 4000 per response for blueprints
- Rate Limits: Tier 1 (60 RPM) for development, Tier 3 (3500 RPM) for production
- Fallback: Cached responses for common queries

6.2 LangChain Configuration

- Version: 0.1.x (stable)
- Chain Type: SequentialChain for multi-agent pipelines
- Output Parsing: StructuredOutputParser with Zod schemas
- Error Handling: Retry with exponential backoff (max 3 retries)
- Logging: All LLM calls logged with tokens used, latency, success/failure

7. Security Requirements

Authentication

OAuth 2.0 + JWT. Clerk SDK integration. Session expiry: 7 days. Refresh token rotation.

Authorization

RBAC: User (own data), Admin (all data), System (service-to-service). Middleware enforcement.

Encryption at Rest

AES-256 for sensitive fields. PostgreSQL native encryption.

Encryption in Transit

TLS 1.3 for all connections. HSTS headers. Secure cookie flags.

Input Validation

Zod schemas for all API inputs. SQL injection prevention via Prisma ORM.

XSS Prevention

React built-in escaping. DOMPurify for HTML content. Content Security Policy headers.

CSRF Protection

SameSite=strict cookies. CSRF tokens for state-changing operations.

API Key Security

OpenAI keys in environment variables only. Server-side proxy. No client exposure.

8. Performance Requirements

Metric	Target	Measurement
Page Load Time (First Contentful Paint)	< 1.5s	Lighthouse
Time to Interactive	< 3s	Lighthouse
API Response Time (GET)	< 200ms	Server logs
API Response Time (POST generate)	< 3s	Server logs
AI Generation Time (ideas)	< 5s	OpenAI API logs
AI Generation Time (blueprint)	< 15s	OpenAI API logs
Database Query Time	< 50ms	Prisma logs
Cache Hit Rate	> 80%	Redis metrics
Uptime	> 99.9%	Uptime monitoring
Concurrent Users	10,000	Load testing

9. Integration Requirements

OpenAI API

REST API. JSON requests/responses. Bearer token auth. Rate limit headers. Error codes: 429 (rate limit), 500 (server error), 503 (overloaded).

GitHub API

REST API v3. OAuth for private repos. 5000 requests/hour for authenticated users. Pagination for large repos.

Clerk Auth

SDK integration. Webhooks for user events. JWT verification. Session management.

Vercel

Git integration. Environment variables. Preview deployments. Analytics API.

Railway/Supabase

PostgreSQL connection string. SSL required. Connection pooling. Backup API.

10. Testing Requirements

Unit Tests

Vitest for React components and utility functions. Coverage target: 70%+. Mock external APIs.

Integration Tests

API route testing with supertest. Database transactions rolled back after each test.

E2E Tests

Playwright for critical user flows: signup → generate → select → export. 90%+ pass rate.

Performance Tests

Lighthouse CI for every PR. Load testing with k6 for 1000 concurrent users.

Security Tests

npm audit for vulnerabilities. Snyk scanning. OWASP ZAP for penetration testing.

Accessibility Tests

axe-core for automated a11y checks. Screen reader testing with NVDA/VoiceOver.

11. Deployment Requirements

CI/CD Pipeline

GitHub Actions: Lint → Test → Build → Deploy. Parallel jobs for speed.

Environments

Development (localhost), Staging (preview URLs), Production (custom domain).

Environment Variables

DATABASE_URL, OPENAI_API_KEY, CLERK_SECRET_KEY, REDIS_URL, etc. Never committed to repo.

Database Migrations

Prisma migrate deploy in CI. Zero-downtime migrations. Rollback plan.

Monitoring

Vercel Analytics (performance). Sentry (errors). UptimeRobot (availability).

Rollback

One-click rollback via Vercel dashboard. Database rollback via backup restore.

12. Maintenance Requirements

Dependency Updates

Weekly npm audit. Monthly dependency updates. Automated PRs via Dependabot.

Database Maintenance

Monthly index optimization. Quarterly vacuum analyze. Annual schema review.

AI Model Updates

Monitor OpenAI API changelog. Test new models in staging. Gradual rollout.

Security Patches

Critical patches within 24 hours. High severity within 1 week. Regular scans.

Backup Verification

Monthly backup restore tests. Documented recovery procedures. RTO < 4 hours.

Performance Review

Monthly performance report. Identify bottlenecks. Optimization sprints quarterly.

— End of TRD —